Resonance Energy Systems

> **Tagline:** *(TBD — see `branding/taglines/TAGLINES_MASTER.md`)*

Document: HB-04 — Hydraulic Ram Pump Design & Tuning

Version: 0.1

Date: 2026-01-22

Applicable contexts: Creek / Canal / Outfall (default)
See `04_Technology/handbooks/_shared/DEFAULT_CONTEXTS.md`

Disclaimer & Positioning

• This handbook is engineering doctrine for deployment and operations.
• Performance claims must be supported by measured head, flow, losses, and efficiency.
• Visionary inspirations may influence design heuristics but do not replace validated physics.

Hydraulic Ram Pump Design & Tuning

Handbook ID: HB-04
Version: 0.1 (Draft)

Deployment Contexts (Default: All Three)

This handbook is written to be valid for three default deployment environments:
• **[CREEK]** Creek / foothills run-of-river (cold + debris)
• **[CANAL]** Irrigation canal drop (high flow + ops crew)
• **[OUTFALL]** Industrial outfall / water-works (24/7 + compliance + corrosion)

Context overlays live in: `contexts/CONTEXT_OVERLAYS.md`.
Shared definitions: `04_Technology/handbooks/_shared/DEFAULT_CONTEXTS.md`.

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1) Overview

(Draft)

2) Key Principles

(Draft)

3) Deployment Contexts (Default: All Three)

(This section will be inserted automatically if missing.)

4) Procedures

(Draft)

5) Failure Modes & Red Flags

(Draft)

6) Checklists

See `checklists/`.

7) References

See `11_References/MASTER_REFERENCES.md`.